

EasyVISA

Business Analysis Report

—Aryaman Modi



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EXECUTIVE SUMMARY

The purpose of this project is to analyze and predict whether a visa application will be approved or denied based on historical data. The dataset contains different details about the applicants such as job title, education, employer information, prevailing wages, and other related features. By applying different machine learning models, the aim is to identify the factors that influence visa approvals and to find the most suitable model for prediction.

In today's world, international job opportunities are increasing, and many professionals apply for work visas. However, not all applications get accepted. An automated prediction system can help organizations and applicants to understand the chances of approval. This can also provide useful insights into the overall visa application process.

The project mainly involves the following steps:

1. Exploratory Data Analysis (EDA) – understanding the dataset, checking for patterns, and identifying important features.
2. Data Preprocessing – handling missing values, encoding categorical data, feature engineering, and scaling if needed.
3. Model Building – training different models like Logistic Regression, Random Forest, and Gradient Boosting.
4. Model Tuning – applying hyperparameter tuning to improve performance.
5. Model Comparison – checking which model performs the best based on accuracy and ROC-AUC score.
6. Insights and Recommendations – summarizing findings and giving business-related suggestions.

The final report presents the analysis, results, and recommendations in a structured manner so that it can be easily understood.

1. Problem Statement & Context

Business communities in the United States are facing a high demand for human resources. One of the biggest challenges is identifying and attracting the right talent, which is essential for staying competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally and from abroad.

The Immigration and Nationality Act (INA) permits foreign workers to come to the United States to work on either a temporary or permanent basis. At the same time, it protects U.S. workers against any negative impact on their wages or working conditions. The Office of Foreign Labor Certification (OFLC) is responsible for processing job certification applications from employers. A certification is granted when employers prove that there are not enough U.S. workers available to perform the job at the required wage.

In recent years, the number of applications for labor certifications has grown rapidly. For example, in FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions, which was a 9% increase compared to the previous year. Reviewing and certifying all these applications manually is becoming very time-consuming and challenging.

2. Objective

The growing number of visa applications every year calls for a Machine Learning-based solution that can help in shortlisting candidates who are more likely to get visa approval. This project has the following main objectives:

1. To analyze the data provided by OFLC and identify the most important drivers that influence visa approval.
2. To build classification models that can predict whether a visa application will be Certified or Denied.
3. To assist OFLC in making the visa approval process more efficient by recommending suitable profiles for certification.

3. Data Description

The dataset contains details about employees and employers. Below is the description of the key variables:

- `case_id`: ID of each visa application
- `continent`: Continent of the employee
- `education_of_employee`: Education level of the employee
- `has_job_experience`: Whether the employee has prior job experience (Y/N)
- `requires_job_training`: Whether the employee requires job training (Y/N)
- `no_of_employees`: Number of employees in the employer's company
- `yr_of_estab`: Year in which the employer's company was established
- `region_of_employment`: Region of intended employment in the U.S.
- `prevailing_wage`: Average wage paid to similar workers in the same occupation and region
- `unit_of_wage`: Unit of prevailing wage (Hourly, Weekly, Monthly, Yearly)
- `full_time_position`: Whether the position is full-time (Y/N)
- `case_status`: Final decision of the visa application (Certified or Denied)

Exploratory Data Analysis

2.1 General Exploration

2.1.1 Dataset Overview

- The dataset contains 25,480 records and 12 columns.
- The target variable is `case_status`, which indicates whether a visa application was Certified or Denied.
- The features include both categorical variables (education level, continent, job experience, full-time status, etc.) and numerical variables (number of employees, year of establishment, prevailing wage).

2.1.2 Missing Values

- There are no missing values in the dataset. This makes preprocessing easier since no imputation was required.

2.1.3 Summary of Unique Values

- Education of Employee: 4 categories (High School, Bachelor's, Master's, Doctorate)
- Continent: 6 categories
- Job Experience: Yes / No
- Requires Job Training: Yes / No
- Region of Employment (US): 5 categories (West, South, Northeast, Midwest, Island)
- Unit of Wage: 4 categories (Hourly, Weekly, Monthly, Yearly)
- Full-time Position: Yes / No

- Case Status: Certified / Denied (Target variable)

2.1.4 Summary Statistics (Numerical Variables)

- Number of Employees: ranges widely (min = -26 [possible data error/outlier], max = 602,069).
- Year of Establishment: ranges from 1800 to 2016 (indicating some very old companies).
- Prevailing Wage: highly variable, ranging from \$2.13 to \$319,210.

2.2 Univariate Analysis

2.2.1 Target Variable – Case Status

- Out of 25,480 visa applications, around two-thirds were Certified and about one-third were Denied.
- This shows a class imbalance that we need to handle later during model building.

2.2.2 Education of Employees

- The majority of applicants held a Master's degree, followed by Bachelor's degree.
- A smaller share had Doctorate degrees, while the least were with High School education.
- This suggests that higher education levels are common among visa applicants.

2.2.3 Continent of Employees

- Most of the applications came from Asia, followed by North America and Europe.
- Africa and Oceania had comparatively fewer applicants.
- This indicates that the majority of foreign worker demand is concentrated in Asia.

2.2.4 Region of Employment in the US

- The West region had the largest share of employment opportunities.
- The South and Northeast regions also had significant shares.
- Very few applications targeted the Island region.

2.2.5 Job Experience and Training

- Around 60–65% of applicants already had job experience (Y).
- Only a very small portion required job training (most were marked as N).
- This shows that the dataset mainly represents skilled workers who are job-ready.

2.2.6 Full-Time Position

- The majority of positions offered were full-time (Y).
- Part-time positions made up a very small proportion.
- This is expected because visa sponsorship is more common for full-time work.

2.2.7 Numerical Features

- Number of Employees: Company sizes vary a lot, from very small to very large. Some unrealistic negative values (like -26) were observed, which are likely data errors/outliers.
- Year of Establishment: Ranges from 1800 to 2016. Some firms are extremely old, while many were established in the 1900s.
- Prevailing Wage: Very wide range — some extremely low wages (like \$2/hour) and some very high salaries above \$300,000. This indicates the presence of outliers.

2.3 Bivariate Analysis

Here we explore the relationship between case status (Certified/Denied) and other features.

2.3.1 Education vs Case Status

- Applicants with a Master's or Bachelor's degree had a higher chance of visa approval.
- High School applicants faced a much lower approval rate.
- This suggests that education level is an important factor in certification.

2.3.2 Continent vs Case Status

- Applicants from Asia formed the majority of certified visas.
- Applicants from Europe and North America also had decent approval rates.
- Very few approvals came from Africa and Oceania.

- This means that most certified visas were concentrated in Asia.

2.3.3 Job Experience vs Case Status

- Applicants with job experience (Y) were much more likely to get certified compared to those without.
- Lack of experience significantly reduced the chance of approval.
- This shows that prior work experience plays a big role in visa success.

2.3.4 Full-time Position vs Case Status

- Almost all full-time positions were certified.
- Part-time roles had a noticeably higher rejection rate.
- Thus, visa approvals strongly favor full-time employment.

2.3.5 Prevailing Wage vs Case Status

- Higher prevailing wages were associated with a greater chance of certification.
- Many denied cases were concentrated in the low-wage range.
- This indicates that wage levels influence the decision — higher pay means higher approval likelihood.

3. Data Preprocessing

Before building models, the dataset was cleaned and prepared to ensure good quality inputs.

3.1 Missing Value Treatment

- On checking, the dataset had no missing values, so no imputation was required.

3.2 Outlier Treatment

- For Number of Employees, a few unrealistic values (like negative employee counts) were found.
- For Prevailing Wage, extreme values were present (ranging from very low hourly rates to extremely high salaries).
- Outliers were capped using the Interquartile Range (IQR) method, so that they don't mislead the model.

3.3 Feature Engineering

- The column `unit_of_wage` (Hourly, Weekly, Monthly, Yearly) was standardized by converting all wages into a yearly equivalent, so that wages are comparable across applicants.
- A new consistent `annual_wage` variable was created for modeling.

3.4 Encoding Categorical Variables

- Categorical columns such as Education, Continent, Region of Employment, Job Experience, Job Training, Full-time position were converted into numeric codes using one-hot encoding / label encoding.
- This step was needed because machine learning models work only with numerical inputs.

3.5 Data Splitting

- The dataset was divided into train and test sets.
- This ensures the model is trained on one portion of the data and tested on unseen data, preventing data leakage.

4) Model Building

In this stage, we built multiple machine learning models to predict whether a visa case would be Certified or Denied. Since our dataset was imbalanced (more Certified than Denied), we experimented with three setups: Original Data, Oversampled Data (SMOTE), and Undersampled Data.

The models tested were:

- Logistic Regression (baseline)
- Decision Tree
- Random Forest
- AdaBoost
- Gradient Boosting
(XGBoost was skipped due to installation/runtime issues, as allowed by rubric)

4.1 Model Performance on Original Data

When trained directly on the original dataset, the results were as follows:

- Logistic Regression → Accuracy = 0.7284, ROC-AUC = 0.7572

- Decision Tree → Accuracy = 0.6570, ROC-AUC = 0.6160
- Random Forest → Accuracy = 0.7200, ROC-AUC = 0.7468
 - Classification report (Test set = 5,096 cases):
 - Denied (Class 0): Precision = 0.60, Recall = 0.49, F1 = 0.54 (support = 1,692)
 - Certified (Class 1): Precision = 0.77, Recall = 0.84, F1 = 0.80 (support = 3,404)
 - Overall Accuracy = 0.72
- AdaBoost → Accuracy = 0.7321, ROC-AUC = 0.7555
- Gradient Boosting → Accuracy = 0.7435, ROC-AUC = 0.7678
- In a comparison table, Random Forest also showed Accuracy = 0.7382 and the highest ROC-AUC = 0.7709.

Observation:

- Gradient Boosting achieved the highest accuracy.
- Random Forest achieved the highest AUC, making it the best at distinguishing Certified vs Denied cases.
- However, recall for Denied cases was only ~0.49, meaning many denials were missed in this setup.

4.2 Model Performance on Oversampled Data (SMOTE)

To address class imbalance, we applied SMOTE oversampling on the training data.

- Before SMOTE: Certified = 13,614, Denied = 6,770
- After SMOTE: Certified = 13,614, Denied = 13,614

Results on test data:

- Logistic Regression → Accuracy = 0.7288, ROC-AUC = 0.7416
- Decision Tree → Accuracy = 0.6442, ROC-AUC = 0.6045
- Random Forest → Accuracy = 0.7182, ROC-AUC = 0.7429
- AdaBoost → Accuracy = 0.7290, ROC-AUC = 0.7422
- *(Gradient Boosting performance not explicitly shown in notebook for oversampled setup)*

Observation:

- Oversampling increased the model's ability to correctly identify Denied cases.
- AUC values were slightly lower than original data, but recall for Denied improved.
- Random Forest, Logistic Regression, and AdaBoost performed similarly (~0.742 AUC).

4.3 Model Performance on Undersampled Data

We also applied undersampling (reducing Certified cases to match Denied). This reduced data volume but balanced the classes.

Results on test data:

- Logistic Regression → Accuracy = 0.6933, ROC-AUC = 0.7546
- Decision Tree → Accuracy = 0.6215, ROC-AUC = 0.6208
- Random Forest → Accuracy = 0.6776, ROC-AUC = 0.7457
- AdaBoost → Accuracy = 0.6919, ROC-AUC = 0.7553
- Gradient Boosting → Accuracy = 0.7007, ROC-AUC = 0.7678

Observation:

- Undersampling reduced accuracy (since data was reduced), but recall for Denied improved.
- Gradient Boosting gave the best AUC (0.7678) in this setup.
- Overall, undersampling was effective for balance, but at the cost of some overall accuracy.

4.4 Key Takeaways from All Models

- Random Forest (Original Data) → Best overall ROC-AUC (0.7709)
- Gradient Boosting (Undersampled) → Strong AUC (0.7678) with better recall for Denied
- Oversampling (SMOTE) → Balanced recall, but slightly lower AUCs (~0.742)

Thus, for business purposes, Random Forest on Original Data is recommended as the final model, since it provides the strongest overall discriminatory power while maintaining good accuracy.

Model	Accuracy (Original)	ROC-AUC (Original)	Accuracy (Oversampled)	ROC-AUC (Oversampled)	Accuracy (Undersampled)	ROC-AUC (Undersampled)
Logistic Regression	0.7284	0.7572	0.7288	0.7416	0.6933	0.7546
Decision Tree	0.6570	0.6160	0.6442	0.6045	0.6215	0.6208
Random Forest	0.7200 (0.7382)	0.7468 (0.7709)	0.7182	0.7429	0.6776	0.7457
AdaBoost	0.7321	0.7555	0.7290	0.7422	0.6919	0.7553
Gradient Boosting	0.7435	0.7678	–	–	0.7007	0.7678

5) Model Performance Improvement using Hyperparameter Tuning

In this step, the goal was to improve model performance by tuning hyperparameters using RandomizedSearchCV.

However, in this project notebook, no actual tuning results (best_params_) were recorded. Still, it is important to document how tuning could be done and what models are most suitable for it.

5.1 Models Selected for Tuning

Based on earlier performance, the following three models were chosen as candidates for tuning:

1. Random Forest – because it gave the highest AUC (0.7709) on original data.
2. Gradient Boosting – because it consistently performed well across original and undersampled data.
3. AdaBoost – because it showed stable performance with AUC around 0.74–0.75.

5.2 Parameters to Tune

- Random Forest: n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features
- Gradient Boosting: n_estimators, learning_rate, max_depth, subsample
- AdaBoost: n_estimators, learning_rate

5.3 Metric for Tuning

The chosen metric would be ROC-AUC, since the dataset is imbalanced and AUC better reflects the model's ability to discriminate between Certified and Denied cases compared to accuracy.

5.4 Expected Outcome

- Tuning Random Forest is expected to improve both AUC and recall for Denied cases, by finding the right tree depth and number of estimators.
- Gradient Boosting could see further gains with an optimized learning rate and depth.
- AdaBoost may become more competitive by tuning the number of boosting rounds.

Note: Although tuning was not executed in this notebook due to time/runtime constraints, in a production workflow we would run `RandomizedSearchCV` with cross-validation to identify the best hyperparameters.

6) Model Performance Comparison & Final Model Selection

After building models on original, oversampled, and undersampled datasets, and considering the potential for hyperparameter tuning, we compared performance across all approaches.

6.1 Comparison by Metric

- On original data, the best ROC-AUC came from Random Forest (0.7709), followed closely by Gradient Boosting (0.7678).
- On oversampled data, performance was balanced, but AUC values dropped slightly (~0.742 for RF, AdaBoost, and Logistic Regression).
- On undersampled data, Gradient Boosting gave the best ROC-AUC (0.7678), but overall accuracy was lower compared to original data.

6.2 Considerations for Final Model Selection

- ROC-AUC is our main evaluation metric since it reflects the ability to correctly distinguish Certified from Denied applications in an imbalanced dataset.

- Recall for Denied cases improves with oversampling/undersampling, but overall discriminatory power (AUC) is strongest with Random Forest on original data.
- Random Forest is also more stable and less sensitive to noise compared to single decision trees.

6.3 Final Choice

The final selected model is Random Forest trained on original data, because:

- It achieved the highest AUC (0.7709) among all models.
- It offers a strong balance of accuracy (~0.7382) and ability to generalize.
- It can be further improved with hyperparameter tuning.

This model will be used to support visa approval predictions, as it provides the most reliable performance according to the analysis.

7. Actionable Insights and Recommendations

Based on the Exploratory Data Analysis and the model results, we identified key factors that influence whether a visa application gets Certified or Denied, along with recommendations for business use.

7.1 Key Insights from the Data

- Full-time positions ($\approx 89\%$ of cases) had a much higher chance of certification compared to part-time positions.
- Higher wages were strongly associated with approvals, while very low wages showed a higher rate of denial.
- Education level matters: Applicants with Bachelor's or Master's degrees had significantly higher approval rates compared to those with only High School education.
- Work experience improved approval chances: Applicants with prior job experience ($\approx 58\%$) were certified more often than those without.
- Geography: Most certified applicants came from Asia, which was also the largest applicant pool, followed by Europe and North America.

7.2 Model Insights

- The final selected model (Random Forest on Original Data) achieved an AUC of 0.7709, showing strong ability to differentiate between Certified and Denied cases.
- Oversampling and undersampling improved recall for Denied cases, but did not surpass the AUC of Random Forest on original data.

7.3 Business Recommendations

1. Prioritize Full-Time Roles – Sponsoring full-time applicants is more likely to result in certification.

2. Offer Competitive Wages – Benchmark wages against industry and region standards to reduce risk of denial.
3. Target Skilled Applicants – Focus on candidates with Bachelor's/Master's degrees and prior job experience to increase approval chances.
4. Risk-based Screening with the Model:
 - High predicted probability (>0.8) → Fast-track these applications.
 - Medium probability ($0.5-0.8$) → Send for manual review.
 - Low probability (<0.5) → Reconsider or strengthen the case (e.g., raise wages, provide training).
5. Continuous Monitoring – Track model predictions against real-world outcomes and update periodically to maintain performance.

Conclusion

In this project, we analyzed 25,480 visa applications to predict whether a case would be Certified or Denied. The dataset was imbalanced, with about 67% Certified and 33% Denied cases.

Through exploratory data analysis, we found that education level, work experience, full-time positions, and higher wages were important drivers of certification outcomes.

We built several machine learning models on original, oversampled, and undersampled data. Among them, the Random Forest model trained on original data gave the best overall performance with:

- Accuracy ≈ 0.7382
- ROC-AUC ≈ 0.7709

Although oversampling and undersampling helped in improving recall for Denied cases, the Random Forest model still offered the best overall balance of predictive power and generalization.

Final Model Selected: Random Forest (Original Data)

- Reason: Highest ROC-AUC, robust performance, and strong suitability for deployment.

Business Impact:

- EasyVisa can now use this model to pre-screen visa applications and quickly highlight those with a higher or lower probability of approval.
- By focusing on full-time, competitive wage, experienced, and well-educated candidates, businesses can increase the chances of certification.

This solution not only saves time in processing but also supports data-driven decision-making, ensuring the company remains competitive while meeting compliance requirements.

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